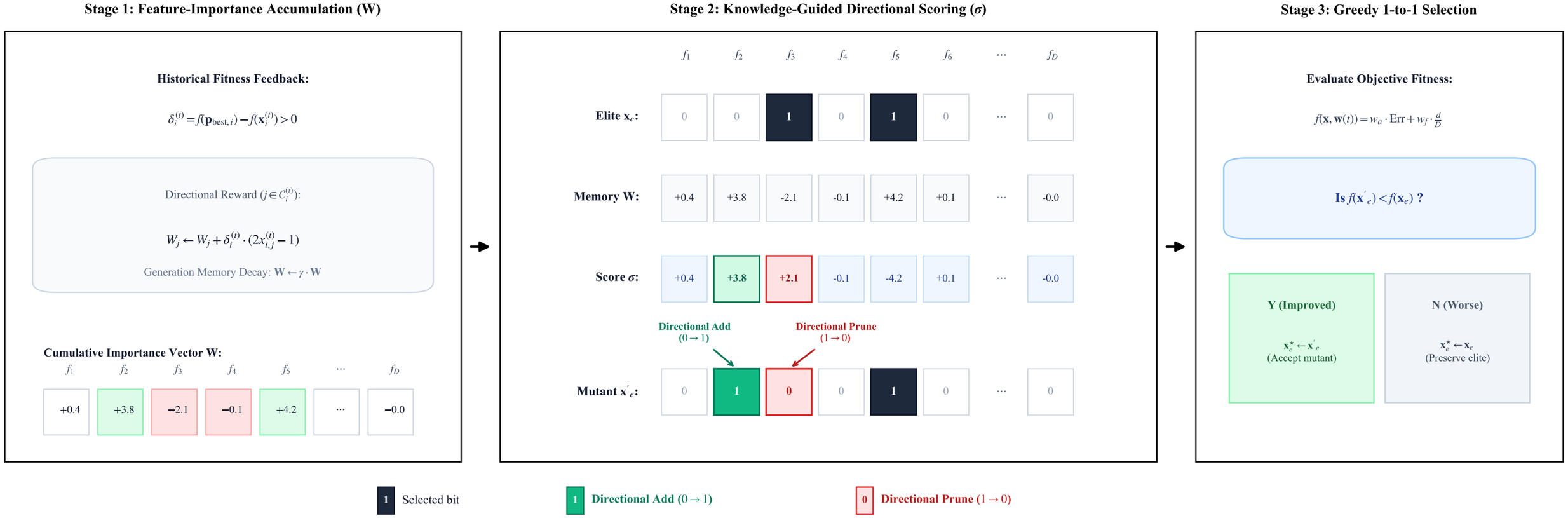

Stage 1: Feature-Importance Accumulation (W)
Historical Fitness Feedback:
$\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}) - f(\mathbf{x}_i^{(t)}) > 0$
Directional Reward ($j \in C_i^{(t)}$):
$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1)$
Generation Memory Decay: $\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}$
Cumulative Importance Vector W:
f_1 f_2 f_3 f_4 f_5 ⋯ f_D
+0.4 +3.8 −2.1 −0.1 +4.2 ⋯ −0.0
Stage 2: Knowledge-Guided Directional Scoring (σ)
f_1 f_2 f_3 f_4 f_5 f_6 ⋯ f_D
Elite $\mathbf{x}_e$: 0 0 1 0 1 0 ⋯ 0
Memory W: +0.4 +3.8 -2.1 -0.1 +4.2 +0.1 ⋯ -0.0
Score σ: +0.4 +3.8 +2.1 -0.1 -4.2 +0.1 ⋯ -0.0
Directional Add (0 → 1)
Directional Prune (1 → 0)
Mutant $\mathbf{x}'_e$: 0 1 0 0 1 0 ⋯ 0
Stage 3: Greedy 1-to-1 Selection
Evaluate Objective Fitness:
$f(\mathbf{x}, \mathbf{w}(t)) = w_a \cdot \text{Err} + w_f \cdot \frac{d}{D}$
Is $f(\mathbf{x}'_e) < f(\mathbf{x}_e)$?
Y (Improved)
$\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$
(Accept mutant)
N (Worse)
$\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$
(Preserve elite)
1 Selected bit
1 Directional Add (0 → 1)
0 Directional Prune (1 → 0)